Bayesian Concentration–Discharge (C–Q) Analysis for Phosphorus

AQUA005 Mini Research Task

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BACKGROUND

Why C–Q Analysis Matters

The Eutrophication Challenge

Phosphorus transport drives water quality degradation in aquatic ecosystems. Understanding how phosphorus concentration (C) responds to discharge (Q) reveals critical transport mechanisms.

Dilution

Decreasing C with increasing Q

Enrichment

Increasing C with increasing Q

Chemostasis

Constant C across flows

Why Bayesian?

Traditional regression methods provide point estimates but fail to capture uncertainty. Bayesian approaches quantify parameter uncertainty through posterior distributions, enabling robust inference about transport mechanisms with explicit confidence intervals.

METHODS

Synthetic Dataset Overview

Data Generation Process

We generated n=200 synthetic observations to test our Bayesian framework under controlled conditions:

- **Discharge:** $\log(Q) \sim \text{Uniform}(-2, 3)$
- **Concentration:** $\log(C) = 0.5 - 0.4 \log(Q) + N(0, 0.3^2)$
- **Correlation:** $\rho = -0.899$ (strong negative)

The synthetic framework embeds known dilution behavior ($\beta = -0.4$), allowing validation of posterior recovery.

	Q	C	logQ	logC
2	0.88047001533197	1.4146927162221474	-0.12729940576318755	0.3469123458317278
3	15.698599931646688	0.5875743406622579	2.753571532049581	-0.5317525036715313
4	5.259151537374817	0.9267549074490328	1.6599697090570253	-0.07606614163320577
5	2.7001097517628074	0.8943819179006083	0.993292420985183	-0.11162239380198355
6	0.2952576843109953	4.70062309088866	-1.2199067977878175	1.5476950724585539
7	0.2952220782245352	3.0961544319403123	-1.2200273983189867	1.130160835601131
8	0.18094142133009145	2.2851729859755423	-1.7095819391590026	0.8264417265028066
9	10.286997600308407	0.7902655830354555	2.330880728874676	-0.2353862089597214
10	2.7334787317118483	0.8231354388798052	1.005575058716044	-0.194634524554614
11	4.6662833065681415	1.1274745561478767	1.5403628889802274	0.11998022553064455
12	0.15000636858992666	4.984881765660786	-1.8970775285209878	1.6064096851106164
13	17.279991315662773	0.4122916054264811	2.8495492608099715	-0.8860243998295018
14	8.690349907503002	0.9269344361544306	2.162213204002109	-0.07587244282754713

MODEL

Bayesian Regression Framework

1

Likelihood Model

The C–Q relationship follows a linear log-log model:

$$log(C) = \alpha + \beta \cdot log(Q) + \varepsilon$$

where $\varepsilon \sim N(0, \sigma^2)$

2

Prior Distributions

Weakly informative priors ensure objectivity:

- $\alpha, \beta \sim N(0, 100^2)$
- $\sigma^2 \sim InvGamma(0.001, 0.001)$

3

Posterior Update

Conjugate normal-inverse-gamma structure enables analytical posterior derivation, avoiding MCMC sampling for this linear model.

The Bayesian approach updates prior beliefs with observed data to produce posterior distributions, quantifying both central estimates and uncertainty for all parameters simultaneously.

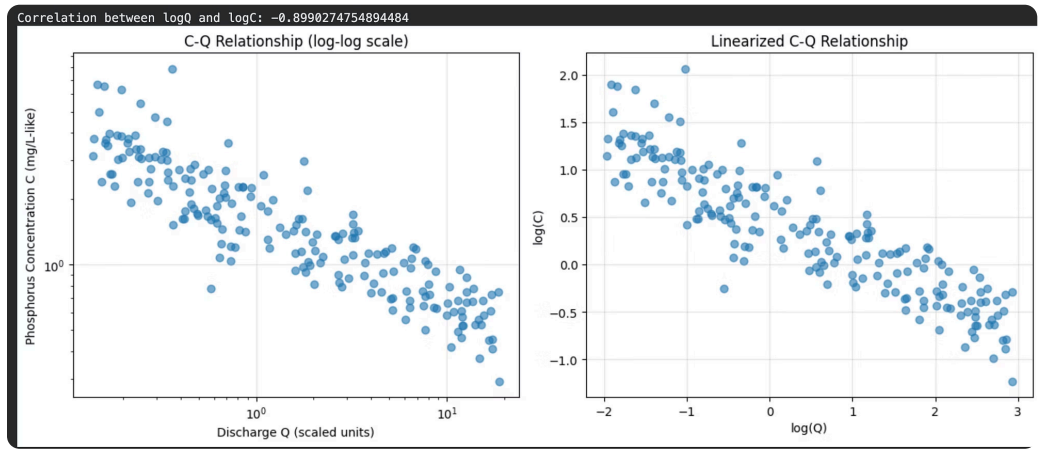
RESULTS

Posterior Parameter Estimates

Parameter	Posterior Mean	95% Credible Interval
Intercept (α)	0.519	[0.478, 0.560]
Slope (β)	–0.402	[–0.430, –0.374]
Std. Dev. (σ)	0.289	—
R^2	0.81	—

Key Findings

- **Excellent recovery:** True β = –0.4, posterior mean = –0.402
- **Narrow credible intervals:** High-precision parameter estimates
- **Strong fit:** R^2 = 0.81 indicates model captures variance



```
=== Data Summary Statistics ===
                Q           C        logQ        logC
count  200.000000  200.000000  200.000000  200.000000
mean     3.857640    1.773272    0.420031    0.352212
std      4.857309    1.270315    1.474457    0.663706
min      0.139124    0.291139   -1.972389   -1.233953
25%      0.424397    0.821539   -0.857088   -0.196581
50%      1.603899    1.392141    0.472431    0.330824
75%      5.955782    2.387275    1.784298    0.870152
max     18.810866    7.876264    2.934435    2.063854
```

INTERPRETATION

Phosphorus Transport Mechanism

Dilution Behavior Confirmed

Negative slope ($\beta = -0.402$) indicates strong dilution: phosphorus concentration decreases as discharge increases.

Magnitude of Effect

Each 10-fold increase in discharge produces approximately 60% reduction in phosphorus concentration ($10^{-0.4} \approx 0.4$).

Process Interpretation

Constant phosphorus loading from sources is progressively diluted during high-flow events. No evidence of enrichment or chemostatic behavior.

The narrow 95% credible interval [–0.430, –0.374] provides high confidence in the dilution mechanism, ruling out alternative transport behaviors.

DISCUSSION

Limitations & Future Directions

Current Limitations

Synthetic Data Only

Analysis based on simulated observations with known parameters for validation purposes.

No Temporal Dynamics

Model assumes stationarity—does not capture seasonal patterns or hysteresis effects.

Single Regime

Unified relationship across all flows may oversimplify baseflow vs. stormflow differences.

Future Research

01

Real-World Application

Apply framework to USGS monitoring data from Vermillion River watershed

02

Regime-Specific Models

Develop hierarchical Bayesian models with separate parameters for baseflow and stormflow

03

MCMC Implementation

Expand to non-conjugate priors and complex likelihood structures using Stan or JAGS

04

Seasonal Analysis

Incorporate time-varying parameters to capture agricultural and climate drivers

Thank You

Questions?

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